DO NOW, AO1 REVISION

Background radiation, different half-lives and uses of nuclear radiation

Atomic structure
Paper 1, Separate Physics only
12 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 3 starred (★) questions first.

	QUESTION	ANSWER	✓
1 ★	**What is background radiation?**	Radiation that is around us all the time, from natural and human-made sources.	☐
2	**Name two natural sources of background radiation.**	Rocks and cosmic rays	☐
3 ★	**Name two human-made sources of background radiation.**	Fallout from nuclear weapons testing and nuclear accidents	☐
4	**What is radiation dose?**	A measure of the risk of harm from exposure to radiation.	☐
5	**Explain why an isotope with a short half-life can be very hazardous at first.**	Its nuclei decay quickly, so its activity is very high.	☐
6	**Give two uses of nuclear radiation in medicine.**	Exploring internal organs, and controlling or destroying unwanted tissue	☐
7	**Explain why alpha radiation is not suitable for exploring internal organs.**	It cannot pass out of the body to be detected, and it is strongly ionising.	☐
8	**State the unit of radiation dose.**	Sievert (Sv)	☐
9	**State how many millisieverts (mSv) equal 1 sievert (Sv).**	1000 mSv	☐
10	**Give two things that affect a person's background radiation dose.**	Their occupation and their location	☐
11 ★	**Explain why a medical tracer should have a short half-life.**	Its activity falls quickly, limiting the dose to the patient.	☐
12	**Explain why a medical tracer should emit gamma radiation.**	It passes out of the body to be detected and is only weakly ionising.	☐

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